HAL open science

Of All StrIPEs: Investigating Structure-informed Positional Encoding for Efficient Music Generation

Manvi Agarwal, Changhong Wang, Gael Richard

To cite this version:

Manvi Agarwal, Changhong Wang, Gael Richard. Of All StrIPEs: Investigating Structure-informed Positional Encoding for Efficient Music Generation. 2026. ⟨hal-05021809v2⟩

HAL Id: hal-05021809

https://hal.science/hal-05021809v2

Preprint submitted on 16 Mar 2026

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.

Distributed under a Creative Commons CC BY-NC-ND 4.0 - Attribution - Non-commercial use - No Derivative Works - International License

Of All StrIPEs: Investigating Structure-informed Positional Encoding for Efficient Music Generation

Manvi Agarwal, *IEEE student member,* Changhong Wang, *IEEE member,* Gaël Richard, *IEEE Fellow.*

***Abstract*—While music remains a challenging domain for generative models like Transformers, a two-pronged approach has recently proved successful: inserting musically-relevant structural information into the positional encoding (PE) module and using kernel approximation techniques based on Random Fourier Features (RFF) to lower the computational cost from quadratic to linear. Yet, it is not clear how such RFF-based efficient PEs compare with those based on rotation matrices, such as Rotary Positional Encoding (RoPE). In this paper, we present a unified framework based on kernel methods to analyze both families of efficient PEs. We use this framework to develop a novel PE method called *RoPEPool*, capable of extracting causal relationships from temporal sequences. Using RFF-based PEs and rotation-based PEs, we demonstrate how seemingly disparate PEs can be jointly studied by considering the interactions they induce between two descriptive levels of the data: the *input*, capturing quickly-varying components, and the *prior*, capturing slowly-varying components. For empirical validation, we use a symbolic music generation task, namely, melody harmonization. We show that RoPEPool, combined with highly-informative structural priors, outperforms all methods.**

***Index Terms*—music generation, symbolic music, transformers, positional encoding, kernels, attention, rotary positional encoding.**

I. Introduction

Generative models have been attracting increasing interest due to their impressive capability to produce high-quality, realistic samples. Among the ingredients necessary for obtaining this superior performance, two play a key role: voluminous data and ever-increasing parameter counts [1]. Domains with large-scale datasets, such as vision, text and speech, use these ingredients to obtain effective large models. In contrast, since music lacks large publicly-available datasets, it is necessary to find new ways to efficiently use existing small-scale datasets.

Another important ingredient is recent architectural advances, such as the development of Transformers and attention. Music has been no stranger to this phenomenon. However, music generated by such models often lacks essential properties that characterize human-generated 'real' music, such as long-term coherence and multi-resolution organization [2]. One simple way of improving music generation is to embed prior knowledge about musical structure into data-driven Transformers [3], [4], [5] through the positional encoding (PE) module [6], [7], [8], [9], which has proven to be a good way to replace the usual (structure-free) PE without an increase in complexity.

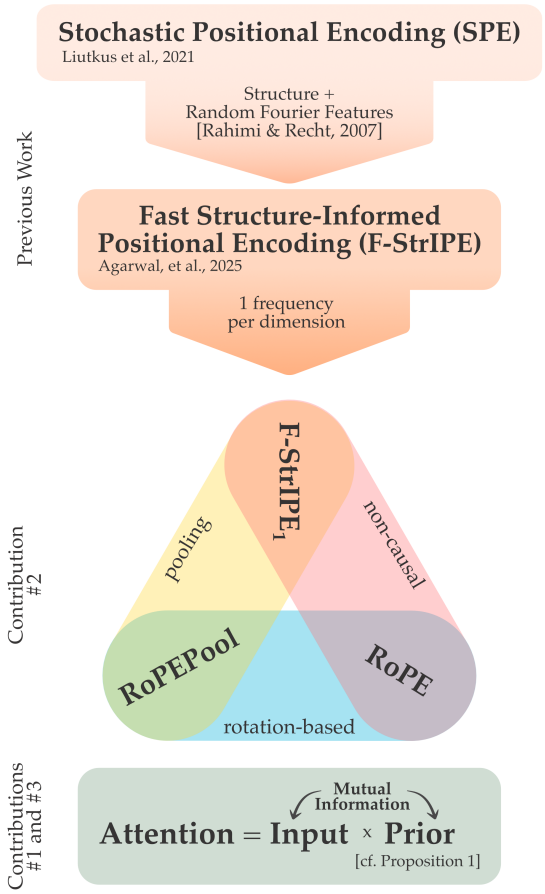

Fig. 1. Top: Connection of previous work to our paper. Middle: Relationship of the three positional encoding methods discussed in this paper (Rotary Positional Encoding or RoPE, Fast Structure-Informed Positional Encoding (1) or F-StrIPE$_1$ and Rotary Positional Encoding with Pooling or RoPEPool). Bottom: Efficient attention enriched with positional information can be viewed as a Tensor Product Kernel operating on two spaces capturing input (quickly-varying, high-frequency) and prior (slowly-varying, low-frequency).

Despite their success, Transformers have a bottleneck: their complexity scales quadratically with sequence length. This can be reduced to a linear cost by using kernel approximations [10], [11]. Recently, low-complexity Transformers were combined with musically-relevant structural information in F-StrIPE [12], which generalizes an older, structure-free PE method, Stochastic Positional Encoding [13], by using Random Fourier Features (RFF) [14]. However, an efficient PE that is conspicuously missing as a benchmark here is Rotary Positional Encoding (RoPE) [15]. Conversely, works that

All authors are based at Information Processing and Communications Laboratory (LTCI), Télécom Paris, Institut Polytechnique de Paris, France.

This work was funded by the European Union (ERC, HI-Audio, 101052978). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

©2026 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works.

investigate RoPE overlook RFF-based efficient PEs, though Stochastic Positional Encoding was the first linear-complexity relative positional encoding.

This reveals a gap in the understanding of efficient positional encoding: how can we comparatively analyze efficient PEs that are based on Random Fourier Features, such as SPE and F-StrIPE, with efficient PEs that are based on rotation matrices, such as RoPE and its variants? A cursory reading of the literature reveals that most works compare PEs largely on a task-dependent basis, with some notable exceptions that also provide insights into the characteristics of different PE methods [16], [17], [18]. We follow the latter line of thought, comparing PEs not just empirically through task performance, but also from a theoretical perspective, using *kernel methods*.

We present three contributions in this paper, which are summarized in Fig. 1.

First, we unify RFF-based PEs and rotation-based PEs into a common framework based on a canonical form of PE-enriched linear attention. We characterize this canonical form as a tensor product kernel operating on two spaces: one encoding similarity in terms of the *input*, capturing high-frequency, quickly-varying components in the data, and the other encoding similarity in terms of the *prior*, capturing low-frequency, slowly-varying components in the data. For rotation-based PEs, we focus on RoPE, while for RFF-based PEs, we construct F-StrIPE$_1$ as an analytical simplification of an existing PE method [12]. By studying the prior-level similarity functions produced by RoPE and F-StrIPE$_1$, we make it possible to compare the forms of attention they induce.

Second, we introduce a novel offshoot of RoPE, called RoPEPool. We first frame PE-enriched attention as feature transforms applied to queries and keys and outline the feature transforms corresponding to RoPE and F-StrIPE$_1$. By doing this, we find that F-StrIPE$_1$ relies on the cross-dimension independence property to use pooling on its feature transform, while RoPE approximates attention by using its feature transform without pooling. We use this insight to combine RoPE's feature transform with the pooling operation to induce cross-dimension interactions, yielding RoPEPool. We use a synthetic dataset to illustrate the rich relationship between the input and the prior in RoPEPool. In particular, we show that, unlike RoPE and F-StrIPE$_1$, RoPEPool is capable of representing both relative or difference-based and absolute or sum-based positional information.

Third, we provide new insights to explain why different types of slowly-varying information in positional encoding lead to better performance. We probe this idea empirically with an application to symbolic music generation. In particular, we use the melody harmonization task to comparatively assess the performance of RoPE, F-StrIPE$_1$ and RoPEPool on different types of priors. We find that RoPEPool with highly-informative priors related to musical structure performs the best on our chosen task. We make our notion of the input-prior interaction precise by computing their mutual information for different prior types. We show that task performance correlates strongly with mutual information and we provide a phenomenological analysis of how the performance of different PE methods changes as the prior carries more information about the input.

The rest of this paper is organized as follows. In Section II, we explain the background literature needed to set the stage for our work. Section III shows how different PEs can be studied using a unified framework of PE-enriched attention as well as present and analyze our novel PE method, RoPEPool. In Section IV, we introduce the experimental protocols we use for empirically validating the ideas we present in Section III. In Sections V and VI, we present the results of our empirical enquiry and show that the concepts developed in Section III are valid for real datasets.

II. Background

In the rest of the paper, we use x for scalars, $\mathbf{x}$ for vectors and $\mathbf{X}$ for matrices. Further, we use the convention "$\mathbf{Q}/\mathbf{K}$" to mean "$\mathbf{Q}$ respectively $\mathbf{K}$".

A. The Role of Positional Encoding in Attention

The Transformer [19] is a sequence-to-sequence architecture. Unlike recurrent neural networks, which perform sequential processing, Transformers process all timesteps in a sequence parallely using a mechanism called *attention*. Given a sequence $[\mathbf{x}_1, ..., \mathbf{x}_T]$ as input, each element of the output sequence is computed as:

$$\mathbf{y}_m = \frac{\sum_n \mathsf{a}_{mn}\mathbf{v}_n}{\sum_n \mathsf{a}_{mn}} \text{ with } \mathsf{a}_{mn} = \exp\left(\frac{a_{mn}}{\sqrt{D}}\right) \tag{1}$$

$$a_{mn} = \mathbf{q}_m\mathbf{k}_n^\top \tag{2}$$

where a_{mn} is the attention coefficient or "similarity score" for a pair of timesteps (m, n). Inputs $\mathbf{x}_m$ are transformed with linear layers to obtain queries $\mathbf{q}_m$, keys $\mathbf{k}_m$ and values $\mathbf{v}_m$. Because attention does parallel processing, it lacks the notion of causality. This makes Transformers invariant to permutations in the temporal order of inputs. Hence, *positional encoding* (PE) is used to provide the model with a sense of time. Positional information can be included in two places: at the input (i.e., via $\mathbf{x}_m$) or during attention computation (i.e., via a_{mn}). Here, we focus on the latter approach, which is called Relative Positional Encoding.

B. Stochastic Positional Encoding: Efficient Attention with Positional Information

The computational cost of attention scales quadratically with sequence length. To bring this cost down to the linear regime, kernelized attention was proposed:

$$\mathsf{a}_{mn} = \mathcal{K}(\mathbf{q}_m, \mathbf{k}_n) = \mathbb{E}\left[\phi(\mathbf{q}_m)\phi(\mathbf{k}_n)^\top\right] \tag{3}$$

where $\mathcal{K}$ is a positive definite kernel and $\phi(\mathbf{x}) : \mathbb{R}^D \to \mathbb{R}^{D_\phi}$ defines a *randomized feature map* for $\mathbf{x}$ [11], [20], [21], [22]. Since it uses multiple random instantiations, ϕ captures, on average, the relationship between $\mathbf{q}_m$ and $\mathbf{k}_n$, typified by $\mathcal{K}$. As a result, coefficients a_{mn} do not need to be computed explicitly, which produces linear-complexity Transformers.

This efficient, kernelized formulation cannot be directly used with Relative Positional Encoding which, as introduced in [23], requires the explicit computation of attention coefficients a_{mn}. This lacuna was filled by Stochastic Positional Encoding

(SPE) [13], which, uses the following *canonical form of PE-enriched exact attention*:

$$a_{mn} = \sum_{d=1}^{D} a_{mn}^{d} = \left[\sum_{d=1}^{D} \mathrm{diag}(\mathbf{Q}_{:,d}) \mathbf{P}_d \mathrm{diag}(\mathbf{K}_{:,d}) \right]_{mn} \tag{4}$$

The positional matrix $\mathbf{P}_d$ captures the relationship between all pairs (m, n) of timesteps that come from the positional index sequences $\mathcal{P}_Q = \{1, ..., m, ..., T_Q\}$, associated with the query matrix $\mathbf{Q}$, and $\mathcal{P}_K = \{1, ..., n, ..., T_K\}$, associated with the key matrix $\mathbf{K}$. In this equation, $\mathbf{Q}_{:,d}/\mathbf{K}_{:,d}$ extracts a T_Q/T_K-dimensional vector containing the dth dimension for all timesteps of the query/key matrix. To obtain approximate attention, SPE factorizes $\mathbf{P}_d$ as:

$$a_{mn} \approx \left[\sum_{d=1}^{D} \overbrace{\mathrm{diag}(\mathbf{Q}_{:,d}) \underbrace{\frac{\tilde{\mathbf{P}}_d^{Q}}{\sqrt{R}} \frac{\tilde{\mathbf{P}}_d^{K\top}}{\sqrt{R}}}_{\approx \mathbf{P}_d}}^{\mathbf{Q}^{\mathrm{SFF}}} \overbrace{\mathrm{diag}(\mathbf{K}_{:,d}^{\top})}^{\mathbf{K}^{\mathrm{SFF}}} \right]_{mn} \tag{5}$$

The matrices $\tilde{\mathbf{P}}_d^{Q}/\tilde{\mathbf{P}}_d^{K}$ contain representations of the positional indices $\mathcal{P}_Q/\mathcal{P}_K$ associated with $\mathbf{Q}/\mathbf{K}$. Similar to the idea behind kernelized attention (3), these matrices collect R instantiations of *Stochastic Fourier Features (SFF)* for $\mathcal{P}_Q$ and $\mathcal{P}_K$, which together provide an approximation of the positional matrix $\mathbf{P}_d$. More precisely, the sequence $\mathcal{P}_Q/\mathcal{P}_K$ is mapped to the matrix $\tilde{\mathbf{P}}_d^{Q}/\tilde{\mathbf{P}}_d^{K}$ as follows:

$$\tilde{\mathbf{P}}_d^{Q/K} = \frac{\mathbf{\Omega}\left(\mathcal{P}_{Q/K}, \boldsymbol{f}_d, \boldsymbol{\theta}_d^{Q/K}\right) \mathrm{diag}\left(\ddot{\boldsymbol{\lambda}}_d\right) \mathbf{Z}_d}{\sqrt{2N_f}} \tag{6}$$

The matrix $\mathbf{\Omega}\left(\mathcal{P}_{Q/K}, \boldsymbol{f}_d, \boldsymbol{\theta}_d^{Q/K}\right)$ contains sinusoidal features for the index sequence $\mathcal{P}_{Q/K}$, parameterized by N_f frequencies (collected in $\boldsymbol{f}_d$) and phase shifts (collected in $\boldsymbol{\theta}_d$). The diagonal matrix $\mathrm{diag}\left(\ddot{\boldsymbol{\lambda}}_d\right)$ is used to apply gains to the sinusoidal features. The matrix $\mathbf{Z}_d$, consists of i.i.d. entries from a zero-mean, unit-variance Gaussian distribution. Seen in this way, we can interpret the matrices $\mathbf{Q}^{\mathrm{SFF}}$ and $\mathbf{K}^{\mathrm{SFF}}$ (5) as PE-enriched versions of $\mathbf{Q}$ and $\mathbf{K}$, combining information about the input ($\mathbf{Q}/\mathbf{K}$) and information about the prior ($\mathcal{P}_Q/\mathcal{P}_K$) by using Stochastic Fourier Features as defined in (6).

As noted earlier [13], because of the properties of $\mathbf{Z}_d$, the cross-dimension terms in (5) are negligible, which allows SPE to go one step beyond (5) by modifying it to:

$$a_{mn} \approx \left[\underbrace{\left(\sum_{d=1}^{D} \mathrm{diag}(\mathbf{Q}_{:,d}) \frac{\tilde{\mathbf{P}}_d^{Q}}{\sqrt{R}} \right)}_{\mathbf{Q}_c^{\mathrm{SFF}}} \underbrace{\left(\sum_{d=1}^{D} \mathrm{diag}(\mathbf{K}_{:,d}) \frac{\tilde{\mathbf{P}}_d^{K}}{\sqrt{R}} \right)^{\top}}_{\mathbf{K}_c^{\mathrm{SFF}}} \right]_{mn} \tag{7}$$

SPE applies the feature maps ϕ, used in kernelized attention (3), to $\mathbf{Q}_c^{\mathrm{SFF}}/\mathbf{K}_c^{\mathrm{SFF}}$ instead of $\mathbf{Q}/\mathbf{K}$. Hence, SPE applies a *sum-pooling* operation to the PE-enriched representations of $\mathbf{Q}$ and $\mathbf{K}$ *before* applying the attention feature maps ϕ to them.

C. Rotary Positional Encoding

Shortly after the development of SPE, an efficient positional encoding method based on rotation matrices called Rotary Positional Encoding (RoPE) [15] was introduced. RoPE approximates PE-enriched attention by applying rotation matrices $\hat{\mathbf{P}}_m^{Q}$ and $\hat{\mathbf{P}}_n^{K}$ to query and key vectors $\mathbf{q}_m$ and $\mathbf{k}_n$:

$$a_{mn} \approx \mathbf{q}_m \left(\hat{\mathbf{P}}_m^{Q}\right)^{\top} \hat{\mathbf{P}}_n^{K} \mathbf{k}_n^{\top} \tag{8}$$

such that the rotations are parameterized by positions m (for the query) and n (for the key). The rotation matrices take a block diagonal form, given as:

$$\hat{\mathbf{P}}_j^{A} = \begin{bmatrix} \mathbf{R}_j^{A}(1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \mathbf{R}_j^{A}(\frac{D}{2}) \end{bmatrix} \tag{9}$$

where each $\mathbf{R}$ is a 2×2 rotation matrix, given as:

$$\hat{\mathbf{R}}_j^{A}(i) = \begin{bmatrix} \cos(f_i \mathcal{P}_A[j]) & -\sin(f_i \mathcal{P}_A[j]) \\ \sin(f_i \mathcal{P}_A[j]) & \cos(f_i \mathcal{P}_A[j]) \end{bmatrix} \tag{10}$$

As hinted in the introduction, RoPE has become tremendously popular, having been deployed in several Large Language Models (LLMs), such as GPT-NeoX [24], Gemma [25], the LLaMa model family [26] and the PaLM model family [27]. With its integration into the PyTorch ecosystem, for example, it has become the PE method of choice for machine learning practitioners working with Transformers. Several works have also dissected and improved on the properties of RoPE, such as its capacity to generalize to unseen sequence lengths [28], [29], [30], [31]. This line of work has lead to insights on its advantages, particularly from a mechanistic interpretability perspective [32].

III. Methods

A. Connecting Stochastic Positional Encoding (SPE) and Random Fourier Features (RFF)

SPE can be directly connected to kernel methods by noting that Random Fourier Features (RFF) [14] are a learning-free, noise-free version of Stochastic Fourier Features (6).

We can see this by first observing that the term $\mathbf{Z}_d \mathbf{Z}_d^{\top}/R = \widehat{\mathbf{C}}_d$ appears in the approximation of $\mathbf{P}_d$ by using Equations (5) and (6). $\widehat{\mathbf{C}}_d$ acts as an empirical covariance matrix and its elements can be written as random variables, with $\alpha = \frac{1}{R}\sum_{i=1}^{R} \upsilon_i^2$ on the diagonal and $\beta = \frac{1}{R}\sum_{i=1}^{R} \upsilon_i \nu_i$ on the off-diagonals, where υ_i and ν_i are standard Gaussian random variables. In the limit of $R \to \infty$, the value of α converges to 1 and the value of β converges to 0, as elaborated in section V of the supplementary materials. As a consequence, the empirical covariance matrix $\widehat{\mathbf{C}}_d$ converges to the theoretical covariance matrix $\mathbf{C}_d = \mathbf{I}_{2N_f}$. In this case, by adjusting the $\frac{1}{2}$ scaling factor coming from (6), the approximation of $\mathbf{P}_d$ simplifies to a weighted mean of phase-shifted cosines:

$$\mathbf{P}_d[m, n] \approx \frac{1}{N_f} \sum_{\omega=1}^{N_f} \Lambda_\omega \cos\left(f_\omega(\mathcal{P}_Q[m] - \mathcal{P}_K[n]) + \Theta_\omega\right) \tag{11}$$

Here, Λ_ω is the gain contributed by the matrices $\mathrm{diag}\left(\ddot{\boldsymbol{\lambda}}_d\right)$ and Θ_ω is the phase-shift contributed by $\boldsymbol{\theta}_d^{Q}$ and $\boldsymbol{\theta}_d^{K}$. By

restricting the gains to be ones, the phase-shifts to be 0, and the frequencies to be drawn i.i.d. from a desired distribution over the frequencies f_ω, we reduce Stochastic Fourier Features to the standard Random Fourier Features.

B. F-StrIPE$_1$: An analytical simplification of Fast, Structure-informed Positional Encoding

Drawing on SPE, a new positional encoding method called Fast, Structure-informed Positional Encoding (F-StrIPE) was proposed for the purpose of efficiently encoding musically-relevant structural information in positional encoding [12].

F-StrIPE reformulates Stochastic Positional Encoding by making two changes. First, it operates in the limit of $R \to \infty$, thereby eliminating the noise introduced by $\mathbf{Z}_d$ as described in Section III-A. Second, it reconceptualizes positional encoding as a flexible way to introduce domain-specific prior knowledge about the underlying data domain of interest. Going beyond positional index sequences (Section II-B), it allows PE to depend on more general index sequences $\mathcal{P} = [p_1, p_2, ..., p_T]$ that can capture different types of information about the input. For instance, it can be designed to convey prior information based on knowledge that we may possess about the data domain, which complements the input information that is already present in the raw data. Standard PE methods, including SPE, typically use time information as the prior, with $p_i = i$. F-StrIPE enables the use of multi-resolution structural labels as an input-dependent prior, with $p_i = \mathbf{s}(i)$, where $\mathbf{s}(i)$ is a vector with L entries describing musical structure. Using this two-step reformulation, F-StrIPE redesigns the feature matrices of (6) by removing $\mathbf{Z}_d$ in the numerator and $\sqrt{2}$ in the denominator as well as replacing time-based index sequences with structure-based index sequences. Thus, F-StrIPE removes the disadvantage of noise from SPE while maintaining its advantage of learnable, data-driven parameters. Broadly, SPE and F-StrIPE can be considered as belonging to the RFF family of PE methods, mirroring the spirit of developments in kernel methods around replacing the Monte Carlo sampling of parameters in traditional RFF with data-dependent, learnable parameters [33].

Although learnable gains and phase-shifts serve to increase the expressivity and empirical performance of F-StrIPE, we recall that RoPE lacks phase-shifts and biases and permits only a single frequency for each dimension in (8)-(10). Keeping in mind that the objective of this work is to compare RoPE with RFF-based PE methods, we choose to restrict the phase-shifts to be 0, the gains to be 1 and $N_f = 1$.This leads to an analytical simplification of F-StrIPE, which we call *F-StrIPE$_1$*. In the rest of this paper, we focus on F-StrIPE$_1$.

C. Interpreting PE-enriched Attention through Kernels

At first glance, F-StrIPE$_1$ and RoPE seem to be completely divergent forms of PE-enriched attention. However, as we will explain in the following sections, they can be jointly analyzed with kernel methods. It is indeed possible to reframe PE-enriched attention in the idiom of kernels by considering the canonical form of PE-enriched exact attention (4) presented in Section II-B, which gives us our first insight:

Proposition 1: Exact attention enriched with positional information (4) can be written as a Tensor Product Kernel [34]:

$$a^d_{mn} = \mathcal{F}(q_{md}, k_{nd}) \times \mathcal{G}_d(p_m, p_n) \tag{12}$$

where $\mathcal{F}$ is the linear kernel $\mathcal{F}(x, x') = x \cdot x'$ applied to q_{md} and k_{nd}, capturing *input-level similarity*, and $\mathcal{G}_d$ is a kernel that depends on the form of $\mathbf{P}_d$ and is applied to indices p_m and p_n, capturing *prior-level similarity*. Changing the form and approximation of $\mathbf{P}_d$ (4) and, consequently $\mathcal{G}_d$ (12) changes the specific form of PE-enriched attention we admit. In fact, $\mathcal{G}_d$ acts as a dimension-specific, input-dependent *mask* on $\mathcal{F}$, which contains the dimension-wise dot-product between the query and the key. Thus, PE-enriched attention is conceptually similar to the causal masking mechanism used in autoregressive Transformers, but it provides a learnable, flexible and fine-grained way of adjusting attention coefficients.

A consequence of Proposition 1 is that we can infer the properties of a_{mn} by decomposing it into its constituents a^d_{mn} and analyzing their properties, specifically by verifying the forms that $\mathcal{F}$ and $\mathcal{G}_d$ take. We provide a sketch proof of Proposition 1 in the supplementary material.

D. Comparing Rotation-based and RFF-based PEs

In this section, we will describe and compare RoPE and F-StrIPE$_1$ in two ways. First, we will characterize the nature of attention (a_{mn}) induced by each of these PE methods by studying the form of $\mathcal{G}_d$ (Proposition 1) they use. Second, we will contrast these methods by reinterpreting PE-enriched attention as feature transforms applied to queries and keys.

For our first goal, we start with a characterization of RoPE through the lens of kernels. We can derive an expression using (8) and (10) for RoPE, which is as follows:

$$a_{mn} \approx \sum_{d=1}^{D/2} \underbrace{(q_{m,2d-1}k_{n,2d-1} + q_{m,2d}k_{n,2d})}_{\mathcal{F}_1} \underbrace{\cos \Delta^-_d}_{\mathcal{G}^{(1)}_d} + \underbrace{(q_{m,2d-1}k_{n,2d} - q_{m,2d}k_{n,2d-1})}_{\mathcal{F}_2} \underbrace{\sin \Delta^-_d}_{\mathcal{G}^{(2)}_d} \tag{13}$$

where we used *pairs of dimensions* as our unit of analysis and we set $\Delta^-_d = f_d(\mathcal{P}_Q[m] - \mathcal{P}_K[n])$. We can see that RoPE captures input-level similarity using $\mathcal{F}_1$ and $\mathcal{F}_2$ and prior-level similarity using $\mathcal{G}^{(1)}_d$ and $\mathcal{G}^{(2)}_d$. Since $\mathcal{F}_2$ is not symmetric in its arguments, $\mathcal{F}_2$ is not a positive definite kernel. As a result, in the case of RoPE, a^d_{mn} and, by extension, a_{mn} are also not positive definite kernels.

For F-StrIPE$_1$, we can write down the equation for attention as:

$$a_{mn} \approx \sum_{d=1}^{D} \underbrace{q_{md}k_{nd}}_{\mathcal{F}} \underbrace{\cos \Delta^-_d}_{\mathcal{G}_d} \tag{14}$$

by using (11) and Section III-B. Since both $\mathcal{F}$ and $\mathcal{G}$ are positive definite kernels in this case, (14) gives us a positive definite kernel for a^d_{mn} and a_{mn}.

Note that, while a generalized framework of linearized relative positional encoding has been previously proposed [35], it fails to make a link between PE-enriched attention and

kernels. For Proposition 1, we used the convention of treating a single dimension d as our unit of analysis, but this can easily be extended to general units of analysis, such as *collections of dimensions*, such that the sum in (4) applies to such collections instead of individual dimensions. Indeed, as we saw in (13) and (14), this is precisely what allows us to unify the analysis of RoPE and F-StrIPE$_1$.

We can further sharpen our understanding of the differences between rotation-based and RFF-based PEs by turning to our second goal and reframing PE-enriched attention as applying feature transforms to the query and key matrices. In this way, we can study the feature transforms induced by RoPE, on the one hand, and F-StrIPE$_1$, on the other.

We begin by focusing on a fact that follows from the construction presented in Section III-B: F-StrIPE$_1$ inherits the pooling operation from SPE. The implication is that, in order for F-StrIPE$_1$ to use the form in (7) for approximating attention, similar to SPE, the cross-dimension terms must be negligible. To understand whether this assumption is valid for F-StrIPE$_1$, we regard (5) and (7) as feature transforms. For (5), we can write the attention matrix $[\mathbf{A}]_{mn} = a_{mn}$ as:

$$\mathbf{A} \approx \left[\mathcal{U}(\mathbf{q}_1) \ \cdots \ \mathcal{U}(\mathbf{q}_{T_Q})\right]^\top \left[\mathcal{U}(\mathbf{k}_1) \ \cdots \ \mathcal{U}(\mathbf{k}_{T_K})\right] \tag{15}$$

where $\mathcal{U}(\cdot)$ is an *unpooled feature transform*, applied to both queries and keys, which can be written as:

$$\mathcal{U}(\mathbf{a}_i) : \mathbf{a}_i \mapsto \left[h(a_{i1}) \oplus h(a_{i2}) \oplus \cdots \oplus h(a_{iD})\right] \tag{16}$$

Here, we used $\mathbf{a}_i$ to stand in for $\mathbf{q}_m$ and $\mathbf{k}_n$ and h as a function that is applied independently to each unit of analysis, which, in this case, is a single dimension. Thus, each dimension of each query/key vector is transformed using $h(\cdot)$ and concatenated together, with $\oplus$ being the concatenation operation, to obtain $\mathcal{U}(\cdot)$. The role of h is to transform PE-free queries and keys into PE-enriched queries and keys, equivalent to the role played by $\mathcal{G}$ with respect to $\mathcal{F}$ (Proposition 1). In contrast to (5), for (7), we can write the attention matrix $\mathbf{A}$ as:

$$\mathbf{A} \approx \left[\mathcal{C}(\mathbf{q}_1) \ \cdots \ \mathcal{C}(\mathbf{q}_{T_Q})\right]\left[\mathcal{C}(\mathbf{k}_1) \ \cdots \ \mathcal{C}(\mathbf{k}_{T_K})\right]^\top \tag{17}$$

where $\mathcal{C}(\cdot)$ is a *sum-pooled feature transform*:

$$\mathcal{C}(\mathbf{a}_i) : \mathbf{a}_i \mapsto \left[h(a_{i1}) + h(a_{i2}) + \cdots + h(a_{iD})\right] \tag{18}$$

where components are added instead of being concatenated.

We can see that $\mathcal{U}(\mathbf{q}_m)\mathcal{U}(\mathbf{k}_n)^\top = \mathcal{C}(\mathbf{q}_m)\mathcal{C}(\mathbf{k}_n)^\top$ only when the *cross-dimension independence property* is fulfilled, i.e., $h(q_{md})h(k_{nd'}) \approx 0$ for $d \neq d'$. In other words, this is precisely when (5) and (7) are equivalent.

Naturally, whether this property is met depends strongly on the nature of h. For F-StrIPE$_1$, h is given by:

$$h(a_{id}) = \left[a_{id}\cos(f_d p_i) \quad a_{id}\sin(f_d p_i)\right] \tag{19}$$

With some simple ideas from harmonic analysis, we can show that, with the assumption of independently sampled frequencies, *the cross-dimension independence property holds for F-StrIPE$_1$*, making it possible to use (7) with F-StrIPE$_1$. A sketch proof is provided in the supplementary materials.

Having described the feature transform for F-StrIPE$_1$, we turn to the feature transform corresponding to RoPE. This has already been described in the literature [15] as:

$$\mathcal{U}(\mathbf{a}_i) : \mathbf{a}_i \mapsto \left[h(a_{i(1:2)}) \oplus \cdots \oplus h(a_{i(D-1:D)})\right] \tag{20}$$

$$h(a_{i(d:d+1)}) = \begin{bmatrix} a_{i,d}\cos(f_d p_i) - a_{i,d+1}\sin(f_d p_i) \\ a_{i,d+1}\cos(f_d p_i) + a_{i,d}\sin(f_d p_i) \end{bmatrix}^\top \tag{21}$$

where h is a function that is now applied independently to pairs of dimensions $(d, d+1)$. Note that $\mathcal{U}(\mathbf{q}_m)\mathcal{U}(\mathbf{k}_n)^\top$ gives us precisely the expected expression for a_{mn} with RoPE (8).

At this point, it is pertinent to observe that SPE and, by extension, F-StrIPE$_1$ use pooling before applying ϕ (7) to approximate downstream linear attention. However, it is possible to directly apply ϕ without pooling while maintaining linear computational complexity by using (5) through its representation as an unpooled feature transform (15). The advantage of working directly with (5) is that it can allow us to use functions h where the cross-dimension independence property is not met. The drawback is that we introduce many more dimensions downstream for approximating linear attention with ϕ. In fact, in contrast with F-StrIPE$_1$, which is able to use (7) and a pooled feature transform for approximating attention, RoPE directly works with (5) and an unpooled feature transform.

To summarize, we showed in this section that RoPE and F-StrIPE$_1$ can be comparatively understood via two perspectives:

1) Using kernel methods: F-StrIPE$_1$ induces a positive definite kernel for attention, whereas RoPE induces a non-positive definite kernel.
2) Rethinking attention as feature transforms: F-StrIPE$_1$ possesses the cross-dimension independence property, which allows pooling before applying ϕ. On the other hand, RoPE does not have this property and directly uses ϕ without pooling.

Perspective (2) also allows us to conclude that, in fact, it is not necessary to apply sum-pooling in order to use downstream linear attention (7). Rather, ϕ can directly be used without sum-pooling (5), as done in RoPE.

E. RoPEPool: Gaining expressivity using simple operations

Since RoPE does not possess the cross-dimension independence property, additional expressivity can be obtained through the cross-dimension interactions by combining RoPE's h function (21) with a sum-pooled feature transform (18) to compute a_{mn}. With this, we can explore how expressive functions h can be leveraged to produce rich input-prior interactions by swapping concatenation for simple operations, such as addition. We then obtain the following expression for attention, analogous to RoPE (13) and F-StrIPE$_1$ (14):

$$\begin{aligned} a_{mn} \approx \sum_{d=1}^{D/2} \Big\{ & (q_{m,2d-1}k_{n,2d-1} + q_{m,2d}k_{n,2d})\cos\Delta^- + \\ & (q_{m,2d-1}k_{n,2d} - q_{m,2d}k_{n,2d-1})\sin\Delta^- + \\ & (q_{m,2d-1}k_{n,2d-1} - q_{m,2d}k_{n,2d})\sin\Delta^+ + \\ & (q_{m,2d-1}k_{n,2d} + q_{m,2d}k_{n,2d-1})\cos\Delta^+ \Big\} \end{aligned} \tag{22}$$

where we used $\Delta^+ = f_d(\mathcal{P}_Q[m] + \mathcal{P}_K[n])$. Similar to our analysis for RoPE and F-StrIPE$_1$, the above equation induces a form of PE-enriched attention that is not a positive definite kernel. Further, in contrast to F-StrIPE$_1$ and RoPE, (22) depends not only on the lag between indices, given by Δ^-, but also their sum, given by Δ^+. Thus, the cross-dimension interactions, obtained by using a pooled feature transform on RoPE's h, lead to a richer representation of prior-level similarity in attention, which now accounts for not just relative relationships through Δ^- but also absolute relationships through Δ^+. Since this new method is obtained by a simple pooling operation on RoPE, we call it *RoPEPool*.

So far, we have focused on distinguishing between different PE methods in rather formal terms, for example, by locating these methods within different kernel families. In the next section, we will combine this with an interpretive approach to compare their expressivities by, first, introducing some simplifications to (13), (14) and (22) to make our comparisons clearer and, second, shedding light on the nature and benefits of prior-level similarity used by RoPEPool.

F. A Simplified Analysis of F-StrIPE$_1$, RoPE and RoPEPool

To study how input and prior interact under different PE methods, we start by making some mild assumptions about the query vectors $\mathbf{q}_m$ and key vectors $\mathbf{k}_n$. We first assume that our basic unit of analysis is 2-dimensional chunks of the query and key vectors, given as $\mathbf{q}_{m,2d-1:2d}$ and $\mathbf{k}_{n,2d-1:2d}$. This does not change anything for RoPE and RoPEPool, but it unifies pairs of dimensions in F-StrIPE$_1$ (14) under a single approximation of $\mathcal{G}_d$. Further, we assume that each 2-dimensional chunk has a Euclidean norm of 1. Then, since we can write the coordinates of any 2D unit-norm vector $\mathbf{v}$ as $[\cos\psi_v, \sin\psi_v]$, where ψ_v is the angle that $\mathbf{v}$ makes with the X-axis, we can think of these chunks solely in terms of their angular representations.

With this, we can first rewrite (13) for RoPE as:

$$a_{mn} \approx \sum_{d=1}^{D/2} \cos\big((\psi_q^d - \psi_k^d) + f_d(\xi_m^d - \xi_n^d)\big) \tag{23}$$

Here, ψ_q^d and ψ_k^d are the angles made by $\mathbf{q}_{m,2d-1:2d}$ and $\mathbf{k}_{n,2d-1:2d}$ with the X-axis, whereas ξ_m^d and ξ_n^d are the angular representations of $\mathcal{P}_Q[m]$ and $\mathcal{P}_K[n]$ in radians. The frequency parameter f_d varies between 0 and 1 and controls the contribution of prior-level similarity to attention.

Similarly, we can rewrite (14) for F-StrIPE$_1$ as:

$$a_{mn} \approx \sum_{d=1}^{D/2} \cos\big(\psi_q^d - \psi_k^d\big)\cos\big(f_d(\xi_m^d - \xi_n^d)\big) \tag{24}$$

and, for RoPEPool, we can rewrite (22) as:

$$a_{mn} \approx \sum_{d=1}^{D/2} 2\cos\big(\psi_q^d + f_d\xi_m^d - \frac{\pi}{4}\big)\cos\big(\psi_k^d + f_d\xi_n^d - \frac{\pi}{4}\big) \tag{25}$$

$$= \sum_{d=1}^{D/2} \cos\big((\psi_q^d + \psi_k^d) + f_d(\xi_m^d + \xi_n^d) - \frac{\pi}{2}\big) + \cos\big((\psi_q^d - \psi_k^d) + f_d(\xi_m^d - \xi_n^d)\big) \tag{26}$$

We give the derivations of (23), (24) and (25) in the supplementary materials. We can comment on the expressivity of PE-enriched attention by studying the relationship between input (ψ_q^d, ψ_k^d) and prior (ξ_m^d, ξ_n^d) in these three equations.

RoPE (23) leads to a linear input-prior interaction. In this case, the prior acts as a *phase-shift* applied to the input. F-StrIPE$_1$ (24) separates input and prior by channeling their interaction through an AND gate-like mechanism. Here, the prior alters the input by acting as a *gain*. A high score is achieved when both the input-level similarity and the prior-level similarity of the query and the key match, i.e., when $(\psi_q^d - \psi_k^d) \approx (\xi_m^d - \xi_n^d)$. Thus, we can see that both RoPE (23) and F-StrIPE$_1$ (24) model the contribution of prior-level similarity to input-level similarity in complementary ways using a single sinusoid.

In contrast, RoPEPool does not just rely on difference-based relationships, with $(\psi_q^d - \psi_k^d)$ and $(\xi_m^d - \xi_n^d)$, but also sum-based relationships, with $(\psi_q^d + \psi_k^d)$ and $(\xi_m^d + \xi_n^d)$. Thus, as shown in (26), RoPEPool uses the superposition of two sinusoids to give richer interaction dynamics between input and prior.

To further clarify these ideas, we simulate a simple toy example with a minimal setting of $D = 2$ for (23), (24) and (25). We devise a synthetic dataset by choosing N prior angles, $C_1, C_2, ..., C_N$, distributed evenly over the interval $(0, \pi)$. Each C_i specifies the mean of a Gaussian distribution $\mathcal{D}_i$, as shown in Fig. 2, where we set $N = 5$. We create a dataset of P points $\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_P\}$, where each $\mathbf{x}_j$ is naturally associated with an angular coordinate ψ_j that has been generated from one of the N Gaussians as $\psi_j \sim \mathcal{N}(C_k, \sigma^2), k \in \{1, ..., N\}$. Further, $\mathbf{x}_j$ is also associated with C_k, the mean of the distribution it was generated from. Thus, for each point $\mathbf{x}_j$, ψ_j defines its *input*, capturing low-level, high-frequency information, and $\xi_j = C_k$ defines its *prior*, capturing high-level, low-frequency information.

We designate one of the P points as our 'query' $\mathbf{x}_q$, which we mark with a star symbol in Fig. 2. We are interested in measuring the score or similarity, given by a_{qj}, between $\mathbf{x}_q$ and all the points in our dataset, using (23), (24) and (25). We show the results in Fig. 3. The query $\mathbf{x}_q$ is marked with a star on the X-axis and we indicate which datapoints share the prior angle of $\mathbf{x}_q$ (marked in blue; we call these *in-group keys* in our discussion below) and which datapoints are generated

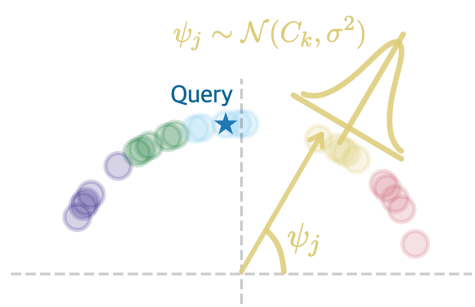

Fig. 2. We generate a toy dataset to study the characteristics of different positional encoding methods for the $D = 2$ setting following the procedure described in Section III-F. Each datapoint (denoted by a circle) is associated with two angles: input ψ_j and prior $\xi_j = C_k$. These angles define the two levels of description for the data: *input* specifies quickly-varying features whereas *prior* defines slowly-varying patterns. We choose a single datapoint as our *query*, marked by $\star$.

from a different prior angle (marked in green; we call these *out-group keys*).

We observe that RoPE and F-StrIPE$_1$ exhibit attention values that are weakly modulated by f and strongly modulated by the input-prior interaction. Discriminability between in-group keys and out-group keys with respect to the query remains robust for the full range of f values. As discussed earlier, high values are achieved only when both the input-level similarity and prior-level similarity of the key match those of the query, which can be seen clearly in Fig. 3. This confirms our analysis of (23) and (24), where both RoPE and F-StrIPE$_1$ capture the input-prior interaction using a single sinusoid. RoPEPool, on the other hand, exhibits an entirely different kind of behaviour. For a fixed value of f, RoPEPool potentiates the scores of the keys in one ψ_j range and depresses the scores of the keys under another ψ_j range. As f changes, these ranges also change. Thus, depending on the value of f, which we set to be learnable, RoPEPool allows the model to dynamically control the keys that will be most similar to a query. This is unlike the static behaviour of RoPE and F-StrIPE$_1$, where a query is most similar to its in-group keys.

The analysis of Fig. 3 can be extended to full attention matrices by selecting each datapoint in our synthetic dataset in turn as the query and computing the attention scores following the same procedure as above. These matrices are shown in Fig. 2 of the supplementary materials for different values of f. As in Fig. 3, RoPE and F-StrIPE$_1$ have similar properties: an increase in f (stronger effect of PE in attention) leads to the emergence of a block-diagonal structure, such that the highest scores become increasingly concentrated in the in-group keys. In contrast, depending on f, RoPEPool can change the keys on which each query focuses. Such attention matrices give us a way to observe how vanilla attention is transformed as the influence of PE increases (i.e., when $f = 0 \rightarrow f = 1$) and provide a concrete illustration of the formulation presented Proposition 1.

Thus, our analysis leads to two conclusions about the comparative expressivities of the three methods discussed. First, RoPEPool is more expressive than RoPE and F-StrIPE$_1$.

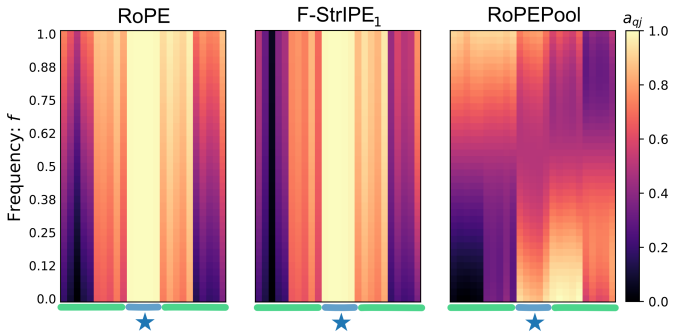

Fig. 3. Using the toy dataset generated with Fig. 2, we calculate attention scores a_{qj} between the query (marked with a $\star$ on the X-axis) and all datapoints for different values of the frequency (marked along the Y-axis). On the X-axis, the range of *in-group keys* is marked in blue and the range of *out-group keys* is marked in green. As we can see from these heatmaps, different PE methods - RoPE (left, using (23)), F-StrIPE$_1$ (middle, using (24)) and RoPEPool (right, using (25)) - exhibit different input-prior trade-offs.

Second, although RoPE and F-StrIPE$_1$ are similar, because F-StrIPE$_1$ allows a nonlinear input-prior interaction, it is more expressive than RoPE.

IV. Experiments

A. Dataset and Input Representation

We use the Chinese POP909 dataset [36] for our experiments. Each MIDI file in this dataset contains 3 tracks: melody, bridge and piano. The bridge acts as a second melody track, supporting the development of the themes in the melody track. The piano track contains polyphonic accompaniment to the melody and bridge tracks. Each file is also paired with fine-grained multi-resolution structural annotations [37]. For this paper, following previous findings [12], we only use chord annotations for structural information, which uses a resolution of quarter-note. We use the POP909 alignment dataset [9] to correctly sync the structural labels with the input. We work with the MIDI data in the form of binary pianorolls $\mathbf{X} \in \mathbb{B}^{(n_{\text{tracks}} \times 128) \times n_{\text{time}}}, \mathbb{B} = \{0, 1\}$, where n_{tracks} is the number of tracks and n_{time} is the number of timesteps in the pianoroll. The POP909 dataset has approximately 400 unique chords. For each chord, we have access to two types of information: the scalar root note and the 12-dimensional chroma vector, which indicates the pitches that make up the chord. We investigate both token-based and vector-based representations of this information, as we will describe in later sections.

B. Task: Melody Harmonization

As input, the model is given the melody and bridge tracks $\left[\mathbf{x}_n \in \mathbb{B}^{(n_{\text{tracks}}-1) \times 128}\right]$, with $n \in \{1, ..., n_{\text{time}}\}$. As output, the model produces all three tracks $\left[\mathbf{y}_n \in \mathbb{B}^{n_{\text{tracks}} \times 128}\right]$. Note that, here, we are in an extreme parallel decoding regime, where we require our model to produce the full accompaniment track at once and do not allow our model to autoregressively condition later predictions on earlier predictions. We train our models on sequences of length 16 bars. We test two settings: the *in-domain* setting, with sequences of length 16 bars, and the *length generalization* setting, with sequences of length 64 bars.

C. Model and Training

We use a 2-layer causal encoder Transformer with 4 heads and 512 model dimension. We train for 15 epochs with a batch size of 8, gradient clipping and curriculum learning [38]. We use two learning rate schedulers: a linear warmup, applied for the first three epochs, and a decay, applied for each epoch. We do a grid-search for two hyperparameters: learning rate (choices: $\{1, 5, 10\} \times 0.0001$) and post-processing binarization strategy [9] (choices: thresholding, thresholding with merge). The first binarization strategy uses a fixed threshold on the logits to obtain binarized outputs. The second binarization strategy, first, uses a fixed threshold to binarize the logits and, second, allows the gap between two binarized notes to be filled when this gap is less than a minimum distance.

D. Types of Prior Information

To understand how different types of priors influence model performance, we represent this information in several ways, varying how positional encoding captures high-level relationships in the input captured.

Because classic positional encoding using time-based index sequences (TIME) is not input-dependent, we hypothesize that it captures *weak* relationships in the input. Since each time-step is assigned a unique token, this representation assumes that all time-steps in a sample are independent.

On the other hand, to make positional encoding flexible and input-dependent as described in Section III-B, we use *binary chroma chord vectors* (BIN), which contain ones for the notes that occur in a chord and zeros for those that don't. This accounts for the harmonic content of each chord, allowing us to identify repetitions of harmonically-similar chords across different samples at the level of the dataset. Our hypothesis is that such information about musical structure captures *strong* relationships in the input.

Between these two extremities, we use two *blended* chord representations, which keep the input-dependent nature of BIN but ablate some prior information. The first is *repetition-based chord tokens* (REP). Here, we first represent each chord as a unique token, taking into account both its root note and chroma vector. Then, to reduce the number of tokens and focus on repetitions within samples, we re-number the tokens in each sample based on their numerical order. The second is *key-matched chord tokens* (KEY). Here, we use the fact that the POP909 dataset provides the key of each song. Using this, we 'normalize' each chord with respect to its song's key, mapping the chord to its counterpart in either the C:major scale or the C:minor scale. For example, the E:minor chord in D:major scale is mapped to the D:minor chord in C:major scale. Thus, we focus on the function of each chord across samples, while blurring harmonic correlations between chords and a song's pitches. Such ablations give us a way to test how changes in prior information affect task performance.

It is worth noting that we selected three hand-designed representations, based in concepts from music theory, to simplify our experiments, but optimization-based representations [39] can also be used with the PE methods studied in this paper.

E. Baselines

To construct baselines for each type of prior information described above, we investigate three variants of RoPE. The first variant, RoPE(A), is the standard method from literature [15], where the frequencies are fixed, following an exponential distribution, and all heads contain the same frequencies. The second variant, RoPE(B), modifies RoPE(A) by allowing different heads to have different frequencies. The third variant, RoPE(C), builds on RoPE(B) by allowing frequencies to be learned as parameters. For each prior type, we choose the RoPE variant that performs the best. Thereafter, we compare these chosen RoPE baselines with F-StrIPE$_1$ and RoPEPool for each context type. For F-StrIPE$_1$ and RoPEPool, we use a system analogous to RoPE(C), where heads have different, learnable frequencies.

F. Objective Evaluation

Following existing work [9], [12], we choose four objective metrics from the literature to compare the ground truth and the generated music. Each focuses on one axis of musical quality.

We use the *Self-Similarity Matrix Distance (SSMD)* [40] to evaluate large-scale and small-scale structural properties. We compute chroma onset vectors, giving the number of onsets per chroma per half-measure, for the target and the prediction. For each chroma onset vector, we calculate its self-similarity matrix (SSM) by taking pairwise cosine similarities between all elements. We take the absolute difference between the target's SSM and the prediction's SSM, giving us a matrix with entries varying between 0 and 2. The SSMD is the mean of this matrix, linearly scaled to vary between 0 and 100.

We use *Chroma Similarity (CS)* [40] to measure melodic consistency. The harmonic match between each half-measure in the target and its counterpart in the prediction is the cosine similarity between their corresponding chroma onset vectors, derived as described above. The mean of these cosine similarities, varying between -1 and 1, is linearly scaled to vary between -100 and 100, and this gives us the CS.

We use *Grooving pattern Similarity (GS)* [2] to assess rhythmic consistency. The grooving pattern of a pianoroll is a vector that encodes a 1 for the quarter-notes where onsets occur and 0 for those where no onsets occur. The XOR function is applied to the grooving patterns of the target and the prediction, after which we take the mean of the resulting vector, giving us a number between 0 and 1. We linearly scale this to vary from 0 to 100, giving us the GS.

To estimate polyphonicity, we use *Note Density Distance (NDD)* [9], [41]. For each 16th-note of the target and prediction, we calculate the total number of pitches that occur. The NDD is the average percentage of missing pitches in the prediction, with the number of pitches in the target giving us the maximum possible value. It varies between 0 and 100.

Note that all four metrics are scale-invariant to sequence length, making the results from the in-domain and length generalization scenarios comparable with each other.

G. Subjective Evaluation

To assess whether our methods produce accompaniments that perceptually pair well with the melody, we conducted a MUSHRA listening test. We recruited 16 participants, ensuring that none of the authors participated in the test. We presented them with a melody track as the reference. We also provided them with melody-accompaniment mixes, where the accompaniment was either the ground truth or produced by using one of the four prior types described in Section IV-D. This gave us a total of 5 stimuli per melody. We asked them to score each stimulus on a scale of 0 to 100 based on "how well the accompaniment goes with the melody". Further details on the design of the listening test are provided in the supplementary materials.[1]

Prior type	Mutual information with data distribution (in nats)
TIME	0.0085
REP	0.0436
KEY	0.0476
BIN	1.0075

TABLE I
MUTUAL INFORMATION BETWEEN INPUT AND PRIOR

V. RESULTS

In Table II, we first present the results on the three variants of RoPE described in Section IV-E. From these, we select the RoPE baseline with the best performance for each prior type and compare this with F-StrIPE$_1$ and RoPEPool. These results are shown in Table III.

For each PE method in both tables, we train models using five seeds and report the mean and standard deviation for each metric. For Table III, we underline the PE method with the best mean score for each prior type. However, we check if there is a statistically significant difference between the underlined method and other methods within the same prior type by using two-sided t-tests [42]. For a pair of PE methods, we first use a Levene test [43] to check whether the pair of score distributions have the same variance. If this is the case, we use Student's t-test to make the comparison, and, if not, we use Welch's t-test [44]. We use a $*$ with the underlined score to indicate that there is a significant difference ($p < 0.05$) between the best (underlined) mean score and the next-best mean score. If we do not find any significant difference in this comparison, we use a † to indicate that there is a statistically significant difference between the best mean score and the third-best mean score. This is to gain a fine-grained understanding on how different PE methods perform relative to each other while using the same prior. In Table III, we additionally also highlight the best PE method among the different underlined PE methods in bold in order to indicate which combination of prior (TIME, REP, KEY, BIN) and PE (RoPE, F-StrIPE$_1$, RoPEPool) is optimal.

A. Ordering Prior Type by Information Content

Before we describe the results in Tables II and III, let us formalize the notion of 'weak relationships' versus 'strong relationships' with regards to prior type, which we introduced in Section IV-D. To do this, we calculate the mutual information (or MI score) between the input distribution and the prior distribution. As elaborated in the supplementary materials, we convert the binary pianorolls into an array of individual pitches and match each pitch with its corresponding index, which changes depending on the prior. Then, we calculate the MI score between the input, represented as pitches, and their indices. We report this score for each prior type in Table I. Intuitively, the mutual information between these two distributions measures how well we can predict one from the other. We expect that a higher MI score makes it easier for the model to learn the data distribution by leveraging prior information via positional encoding.

[1]Demos can be found at this link

B. Comparing PE Methods within Prior Types

For each prior type, we first describe the results for selecting a RoPE baseline. Then, we compare the results of the chosen RoPE baseline with those of the other PE methods.

For Table II, in the in-domain setting, we see statistically significant results for two prior types, namely, REP and BIN. In the length generalization setting, we see this for three prior types, namely TIME, REP and BIN. For REP, RoPE(B) performs the best and, for BIN, RoPE(C) performs the best, in both settings. For TIME, RoPE(C) performs the best, particularly in the length generalization setting. Finally, for KEY, there is no statistically significant difference between the different PE methods, with RoPE(B) or RoPE(C) achieving the best mean score in most metrics for both settings.

Similar to Table II, TIME in Table III does not have statistically significant results in the in-domain scenario, but has statistically significant results in the length generalization scenario. Similarly, for KEY, F-StrIPE$_1$ and RoPEPool jointly perform best, although, unlike Table II, the difference with RoPE is statistically significant. In REP, RoPE(B) continues its win from Table II and outperforms the other PE methods. For BIN, RoPEPool performs the best, with F-StrIPE$_1$ showing similar performance with a slightly worse mean score.

C. Utility of Structural Information

In Table III, on comparing the best PE methods from each prior type, we find that BIN, encoding global musically-relevant information, performs the best, and TIME performs the worst, with the blended representations, namely, REP and KEY, occupying the middle ranks. In both the in-domain and length generalization scenarios, KEY enables higher performance than REP, although the difference between the two is not significant for GS in the in-domain setting ($p > 0.5$).

D. Length Generalization

In Table III, with the exception of RoPE applied to REP, all methods and prior types lead to an amelioration of scores from the in-domain scenario to the length generalization scenario, which is particularly dramatic for GS in BIN. Testing on out-of-domain samples tends to augment the score differences between various PE methods, especially for GS.

E. Subjective Evaluation

We present the results of our listening test in Fig. 4. We show the mean score as a circle along with the standard deviation with blue error bars and the standard error of the mean with black error bars. To assess whether each prior type is significantly different from the target and the other prior types, we also computed the p-values for all pairwise comparisons for each sample type from Fig. 4 using the methodology described earlier in Section V. We observed that all prior types are significantly different from the target as well as each other with p-values being less than 0.001, except for the comparison between KEY and REP, where the p-value is 0.002.

Configuration	Train = 16 bars; Test = 16 bars				Train = 16 bars; Test = 64 bars			
	CS ↑	SSMD ↓	GS ↑	NDD ↓	CS ↑	SSMD ↓	GS ↑	NDD ↓
Time (TIME)								
RoPE(A)	2.44 ± 0.06	14.65 ± 0.0	7.46 ± 0.05	94.1 ± 0.04	5.83 ± 3.27	13.8 ± 0.0	18.27 ± 9.1	91.79 ± 0.84
RoPE(B)	1.61 ± 0.74	14.66 ± 0.01	6.72 ± 0.71	94.38 ± 0.27	2.23 ± 0.09	13.8 ± 0.0	8.44 ± 0.21	92.72 ± 0.03
RoPE(C)	4.5 ± 4.16	14.65 ± 0.02	10.42 ± 5.32	93.82 ± 1.01	9.1 ± 0.49†	13.79 ± 0.0*	26.78 ± 1.19†	91.26 ± 0.2†
Repetition-based chord tokens (REP)								
RoPE(A)	5.42 ± 5.51	14.61 ± 0.06	10.89 ± 6.65	92.75 ± 2.27	15.42 ± 1.18	13.67 ± 0.01	42.4 ± 3.23	86.77 ± 0.92
RoPE(B)	18.37 ± 1.11*	14.36 ± 0.03*	25.63 ± 1.76*	86.13 ± 0.57*	18.17 ± 1.92*	13.46 ± 0.02*	49.56 ± 3.12*	82.9 ± 0.54*
RoPE(C)	4.59 ± 5.16	14.59 ± 0.11	9.76 ± 5.6	92.84 ± 2.57	9.46 ± 7.28	13.62 ± 0.15	27.27 ± 20.17	88.35 ± 4.06
Key-matched chord tokens (KEY)								
RoPE(A)	11.05 ± 5.62	14.49 ± 0.09	16.89 ± 6.59	89.64 ± 2.56	14.59 ± 5.24	13.63 ± 0.08	39.02 ± 12.34	86.82 ± 2.35
RoPE(B)	11.85 ± 4.6	14.5 ± 0.07	18.49 ± 5.52	89.85 ± 1.99	16.2 ± 2.21	13.63 ± 0.01	43.67 ± 3.95	85.87 ± 0.39
RoPE(C)	12.03 ± 4.51	14.49 ± 0.07	18.57 ± 5.37	89.87 ± 1.93	15.84 ± 1.09	13.62 ± 0.02	42.04 ± 2.65	86.68 ± 0.85
Binary chroma vectors (BIN)								
RoPE(A)	1.81 ± 0.34	14.64 ± 0.01	6.76 ± 0.35	94.3 ± 0.14	10.04 ± 4.21	13.74 ± 0.02	30.39 ± 11.72	89.44 ± 1.69
RoPE(B)	1.9 ± 0.18	14.66 ± 0.01	7.07 ± 0.18	94.29 ± 0.07	3.51 ± 2.56	13.8 ± 0.0	11.93 ± 7.12	92.31 ± 0.73
RoPE(C)	22.14 ± 0.56*	13.98 ± 0.01*	26.45 ± 0.52*	79.25 ± 0.29*	26.27 ± 0.44*	13.13 ± 0.01*	66.81 ± 0.6*	75.02 ± 0.34*

TABLE II

COMPARING DIFFERENT ROTARY POSITIONAL ENCODING (ROPE) BASELINES (SECTION IV-E) USING DIFFERENT PRIORS (SECTION IV-D)

Configuration	Train = 16 bars; Test = 16 bars				Train = 16 bars; Test = 64 bars			
	CS ↑	SSMD ↓	GS ↑	NDD ↓	CS ↑	SSMD ↓	GS ↑	NDD ↓
Time (TIME)								
RoPE(C)	4.5 ± 4.16	14.65 ± 0.02	10.42 ± 5.32	93.82 ± 1.01	9.1 ± 0.49*	13.79 ± 0.0	26.78 ± 1.19*	91.26 ± 0.2
F-StrIPE$_1$	6.01 ± 2.67	14.64 ± 0.02	11.9 ± 3.24	93.2 ± 0.76	6.68 ± 1.12	13.79 ± 0.0	20.67 ± 2.59	91.4 ± 0.21
RoPEPool	2.49 ± 2.64	14.66 ± 0.01	7.79 ± 3.32	94.28 ± 0.68	7.23 ± 0.62	13.79 ± 0.0*	22.13 ± 1.65	91.18 ± 0.2
Repetition-based chord tokens (REP)								
RoPE(B)	18.37 ± 1.11†	14.36 ± 0.03†	25.63 ± 1.76†	86.13 ± 0.57†	18.17 ± 1.92*	13.46 ± 0.02*	49.56 ± 3.12*	82.9 ± 0.54*
F-StrIPE$_1$	17.71 ± 0.97	14.37 ± 0.03	24.82 ± 1.22	86.81 ± 0.71	13.85 ± 1.18	13.59 ± 0.03	37.66 ± 1.93	86.47 ± 0.66
RoPEPool	14.23 ± 1.13	14.47 ± 0.02	20.87 ± 1.86	88.81 ± 0.29	11.67 ± 0.58	13.7 ± 0.01	34.11 ± 1.12	88.04 ± 0.52
Key-matched chord tokens (KEY)								
RoPE(C)	12.03 ± 4.51	14.49 ± 0.07	18.57 ± 5.37	89.87 ± 1.93	15.84 ± 1.09	13.62 ± 0.02	42.04 ± 2.65	86.68 ± 0.85
F-StrIPE$_1$	19.73 ± 0.4	14.1 ± 0.02*	24.08 ± 0.48	81.41 ± 0.27†	23.6 ± 0.51†	13.2 ± 0.05†	60.87 ± 1.28†	77.47 ± 0.35†
RoPEPool	21.05 ± 1.22†	14.18 ± 0.06	26.05 ± 1.53*	82.46 ± 0.96	23.11 ± 1.43	13.28 ± 0.09	59.74 ± 2.46	78.69 ± 1.37
Binary chroma vectors (BIN)								
RoPE(C)	22.14 ± 0.56	13.98 ± 0.01	26.45 ± 0.52	79.25 ± 0.29	26.27 ± 0.44	13.13 ± 0.01	66.81 ± 0.6	75.02 ± 0.34
F-StrIPE$_1$	24.91 ± 0.35	**13.73 ± 0.03**†	28.19 ± 0.31	77.15 ± 0.33	28.3 ± 0.81	**12.95 ± 0.04**†	71.35 ± 0.99	73.97 ± 0.48
RoPEPool	**25.83 ± 1.02**†	13.74 ± 0.07	**29.11 ± 0.97**†	**76.73 ± 1.09**†	**28.6 ± 0.8**†	12.99 ± 0.06	**72.56 ± 1.54**†	**73.77 ± 1.23**

TABLE III

COMPARING DIFFERENT POSITIONAL ENCODING METHODS USING DIFFERENT PRIORS. WE CHOSE THE BEST ROTARY POSITIONAL ENCODING (ROPE) BASELINE FROM TABLE II TO COMPARE AGAINST FAST STRUCTURE-INFORMED POSITIONAL ENCODING (1) (F-STRIPE$_1$) AND ROTARY POSITIONAL ENCODING WITH POOLING (ROPEPOOL)

VI. DISCUSSION

A. Explaining the Success of Structural Positional Encoding with Mutual Information

Table I formalizes our intuition that TIME contains little information about the input distribution, since it assumes that all time-steps within a sample are independent, while REP, KEY and BIN introduce progressively stronger prior information about the input distribution. Note that even though KEY has a higher MI score than REP, they both fall within the same order of magnitude. We contrast this with the comparison between TIME, REP/KEY and BIN, which lead to MI scores at increasing orders of magnitude. This captures the intuition that moving from weak to blended to strong relationships, by varying prior types, leads to fundamental changes in what additional insight can be gained with respect to the input distribution, whereas moving within the same informational regime (e.g., REP to KEY) introduces subtler interactions.

By combining our observations from Sections V-A and V-C, we argue that this is precisely what makes task-specific priors, such as structural information, better than generic priors, such as time information, in positional encoding. This point is further strengthened by the fact that the ranking in MI scores (Table I) matches precisely the ranking in performance with respect to prior type (Table III). These findings could also explain how and why harnessing information about the structural properties of data has been found to aid learnability and improve performance in other generative architectures such as diffusion models, both with synthetic [45] and real data [46].

Having translated our intuitions into a quantitative form, we can now analyze the empirical results from Tables II and III in further detail while bearing these ideas in mind.

B. A Phenomenological Perspective on Performance vis-á-vis Information Content

As presented in Section V-B, the impact of prior type on performance has two facets: the relative performance of different PE methods (i.e., which method performs best) and the

difference between these methods (i.e., is the best performance significantly different from the others). To disentangle them, we can plot the in-domain performance of each PE method on each metric from Table III against the MI scores from Table I. The results are displayed in Fig. 5.

First, this plot, along with Table III, suggests the existence of a transition from no differentiation between PE methods, when only weak relationships are captured under TIME, to simpler PE methods performing better than more complex PE methods under REP and KEY, to more complex PE methods performing better than simpler PE methods, when strong relationships are captured under BIN. Thus, as PE starts to be more informative about the input, there is also

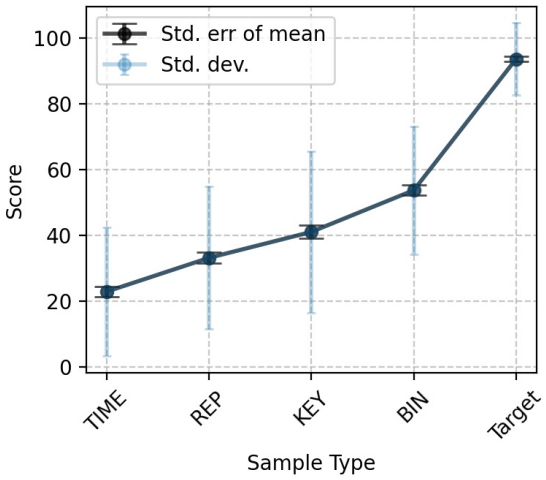

Fig. 4. Scores of listening test averaged over all participants and all trials displaying mean (circle), standard error of the mean (black error bars) and standard deviation (blue error bars).

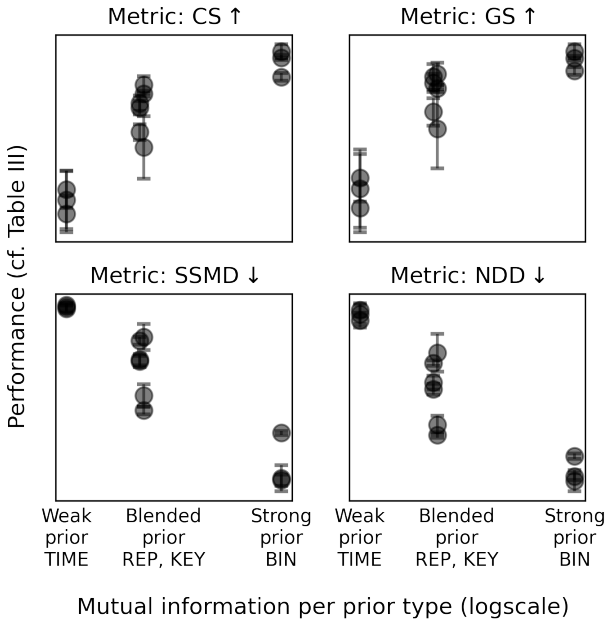

Fig. 5. Performance on objective metrics as a function of how much information about the input distribution is contained in the prior distribution; drawn from the in-domain results in Table III

the *emergence of differentiation* between PE methods. We can see this in Fig. 5 by comparing TIME, where the different PE methods largely overlap, with other prior types, where the PE methods develop distinct distributions. This helps us to give a more fine-grained view of the intuition we developed from the toy example in Section III-F, where RoPEPool emerged as the most expressive and complex of the three PE methods analyzed. In fact, the toy example corresponds to the scenario of a perfect MI score because, in Section III-F, we used the ground-truth prior information in PE, leading to high predictability between the input and the prior. Matching our hypothesis, RoPEPool performs the best under BIN in Table III. However, we think that chord information still does not fully characterize the underlying distribution that generates the desired musical signal, which explains why RoPEPool and F-StrIPE$_1$ do not show significant differences under BIN in Table III. This hints that, as the prior continues to encode more information about the input, even beyond chord arrangements, the comparative task performances of RoPE, F-StrIPE$_1$ and RoPEPool will converge to our hypothesis on expressivity ranking, as proposed in Section III-F.

Second, Table III and Fig. 5 show that the mutual information between data and its PE representation correlates strongly with performance, adding further evidence to our analysis in Section VI-A. This conclusion is also borne out by plotting the mean perceptual scores from Fig. 4 against the MI scores from I, as shown in Fig. 3 of the supplementary materials. Therefore, the more information that is encoded about the input distribution with positional encoding, the better is the performance of the model, in terms of not just objective metrics but also perceptual quality.

This conclusion is further supported by the observation that REP, KEY and BIN outperform TIME on several Music Structure Analysis metrics (i.e. true-to-guess, guess-to-true scores, precision, recall, and F-score for the match between predicted and target segment boundaries) at both fine and coarse resolutions, as evidenced in Section VIII of the supplementary materials. Thus, REP, KEY and BIN are able to reproduce the small-scale and large-scale properties of the target better than TIME.

Finally, Table III and Fig. 5 show that imperfect mutual information (REP/KEY) leads to a large spread of results. This region of positional information is defined by large error bars and a wide range of mean scores.

These observations enable us to characterize the interaction between prior type and comparative performance of PEs using two interrelated phenomena. First, increasing the relevance of the prior distribution with respect to the input distribution diversifies performance, such that differences between PEs become more evident as the prior starts to encode more information about the input. This indicates that empirical works comparing different PE methods might see none or highly dampened differences in the performance of different PEs, not due to the PE's inherent properties, but because the PE does not encode enough information about the input, for example, in the typical case where TIME is used. Second, there exists an intermediary stage of relevance that exhibits a large variance in performance among the different PE methods. This indicates

that, under imperfect structural information, performance is highly initialization-dependent and optimization in such cases might be unstable. This might also explain why, for example, on REP, RoPE(B) does better than RoPE(C) in Table II and better than F-StrIPE$_1$ and RoPEPool in Table III.

While strengthening prior information about musical structure in positional encoding helps performance, we still observe a gap between predictions and targets. This is evident from NDD (Table III) and the gap in scores between the target and all PE methods on the listening test (Fig. 4). Nevertheless, our open-ended subjective test design allows us to conclude that, compared to the different types of structural information provided to the model, human evaluators significantly prefer BIN to the other methods. Hence, although BIN does not reproduce the target perfectly, it produces accompaniments that pair well with the melody. Further, the preference order revealed in the listening test - namely, TIME, REP, KEY and BIN from least to most preferred - matches our objective analysis and lends perceptual grounding to our conclusions.

VII. Conclusion

In this paper, we compared two families of efficient positional encoding methods: those based on Random Fourier Features and those based on rotation matrices. We began by unifying these families under a common framework of PE-enriched attention, which we also used to develop a novel PE called RoPEPool. With a toy example, we studied the expressiveness of RoPE, F-StrIPE$_1$ and RoPEPool. We found that, on the one hand, we can model attention in RoPE and F-StrIPE$_1$ using a single sinusoid. In RoPE, positional encoding modulates vanilla attention using phase, whereas F-StrIPE$_1$ does this using gain. On the other hand, we showed that RoPEPool corresponds to a superposition of two sinusoids and exhibits richer input-prior interaction dynamics than RoPE and F-StrIPE$_1$. To assess the validity of our ideas, we used melody harmonization, a symbolic music generation task. We also varied the type and representation of the index sequences used in PE, ordering them by how much they tell us about the structure of the data. We found that combining RoPEPool with highly-informative prior information about musical structure performs the best, matching our conclusions from the toy example. We went beyond these task-specific findings to clarify the role of information about data structure in PE. We used the mutual information between the input distribution and the prior distribution (via the PE) to explain why rich priors aid learnability. We provided a phenomenological account of how the performances of different PEs vary as priors convey more information about the input, enriching our comparative study. In future work, we aim to expand our analytical understanding from F-StrIPE$_1$, using one frequency per dimension, to F-StrIPE, using multiple frequencies, gains and phases per dimension. In particular, a promising line of research would investigate how better approximations of $\mathbf{P}_d$ can lead to improvements on empirical performance in real-world tasks.

Acknowledgements

We thank Emmanouil Benetos for helpful discussions around variations of chord representations.

References

[1] J. Kaplan, S. McCandlish, et al., "Scaling laws for neural language models," *arXiv*, 2020. [Online]. Available: https://arxiv.org/abs/2001.08361

[2] S. Wu and Y. Yang, "The Jazz Transformer on the Front Line: Exploring the Shortcomings of AI-composed Music through Quantitative Measures," *Conference of the International Society for Music Information Retrieval (ISMIR)*, 2020.

[3] S. Ji, X. Yang, et al., "A survey on deep learning for symbolic music generation: Representations, algorithms, evaluations, and challenges," *ACM Computing Surveys*, vol. 56, no. 1, 2023. DOI: 10.1145/3597493

[4] G. Richard, V. Lostanlen, et al., "Model-based deep learning for music information research: Leveraging diverse knowledge sources to enhance explainability, controllability, and resource efficiency," *IEEE Signal Processing Magazine*, vol. 41, no. 6, pp. 51–59, 2024. DOI: 10.1109/MSP.2024.3415569

[5] K. Bhandari and S. Colton, "Motifs, Phrases, and Beyond: The Modelling of Structure in Symbolic Music Generation," *International Conference on Computational Intelligence in Music, Sound, Art and Design*, 2024.

[6] Y. Ren, J. He, et al., "PopMAG: Pop Music Accompaniment Generation," *ACM International Conference on Multimedia (MM)*, 2020.

[7] J. Liu, Y. Dong, et al., "Symphony Generation with Permutation Invariant Language Model," *Conference of the International Society for Music Information Retrieval (ISMIR)*, 2022.

[8] Z. Guo, J. Kang, et al., "A Domain-Knowledge-Inspired Music Embedding Space and a Novel Attention Mechanism for Symbolic Music Modeling," *AAAI Conference on Artificial Intelligence (AAAI)*, 2023.

[9] M. Agarwal, C. Wang, et al., "Structure-Informed Positional Encoding for Music Generation," *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[10] Y. Tay, M. Dehghani, et al., "Efficient transformers: A survey," *ACM Computing Surveys*, vol. 55, no. 6, 2022.

[11] Y.-H. H. Tsai, S. Bai, et al., "Transformer Dissection: An Unified Understanding for Transformer's Attention via the Lens of Kernel," *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

[12] M. Agarwal, C. Wang, et al., "F-stripe: Fast structure-informed positional encoding for symbolic music generation," *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

[13] A. Liutkus, O. Cífka, et al., "Relative Positional Encoding for Transformers with Linear Complexity," *International Conference on Machine Learning (ICML)*, 2021.

[14] A. Rahimi and B. Recht, "Random features for large-scale kernel machines," *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2007.

[15] J. Su, M. Ahmed, et al., "Roformer: Enhanced transformer with rotary position embedding," *Neurocomputing*, vol. 568, p. 127 063, 2024. DOI: https://doi.org/10.1016/j.neucom.2023.127063

[16] G. Ke, D. He, et al., "Rethinking positional encoding in language pre-training," *International Conference on Learning Representations (ICLR)*, 2021.

[17] T. Chang, Y. Xu, et al., "Convolutions and self-attention: Re-interpreting relative positions in pre-trained language models," *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.

[18] P.-C. Chen, H. Tsai, et al., "A simple and effective positional encoding for transformers," *Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

[19] A. Vaswani, N. Shazeer, et al., "Attention Is All You Need," *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2017.

[20] K. M. Choromanski, V. Likhosherstov, et al., "Rethinking Attention with Performers," *International Conference on Machine Learning (ICML)*, 2021.

[21] A. Katharopoulos, A. Vyas, et al., "Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention," *International Conference on Machine Learning (ICML)*, 2020.

[22] S. Zhuoran, Z. Mingyuan, et al., "Efficient Attention: Attention with Linear Complexities," *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021.

[23] P. Shaw, J. Uszkoreit, et al., "Self-attention with relative position representations," *Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2018.

[24] S. Black, S. Biderman, et al., "GPT-NeoX-20B: An open-source autoregressive language model," *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

[25] G. Team, T. Mesnard, et al., *Gemma: Open models based on gemini research and technology*, 2024. [Online]. Available: https://arxiv.org/abs/2403.08295

[26] H. Touvron, T. Lavril, et al., "Llama: Open and efficient foundation language models," 2023. arXiv: 2302.13971 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2302.13971

[27] A. Chowdhery, S. Narang, et al., "Palm: Scaling language modeling with pathways," *J. Mach. Learn. Res.*, vol. 24, no. 1, Jan. 2023, ISSN: 1532-4435.

[28] kaikondev, *Things i'm learning while training superhot*, 2023. [Online]. Available: https://kaiokendev.github.io/til

[29] S. Chen, S. Wong, et al., "Extending context window of large language models via positional interpolation," *arXiv*, 2023. [Online]. Available: https://arxiv.org/abs/2306.15595

[30] S. Wang, I. Kobyzev, et al., "Resonance rope: Improving context length generalization of large language models," *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[31] S. Pawar, S. M. T. I. Tonmoy, et al., "The what, why, and how of context length extension techniques in large language models – a detailed survey," *arXiv*, 2024. [Online]. Available: https://arxiv.org/abs/2401.07872

[32] F. Barbero, A. Vitvitskyi, et al., "Round and round we go! what makes rotary positional encodings useful?" *International Conference on Learning Representations (ICLR)*, 2025.

[33] Y. Li, K. Zhang, et al., "Learning adaptive random features," *AAAI Conference on Artificial Intelligence (AAAI)*, 2019.

[34] B. Scholkopf and A. J. Smola, *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, 2001.

[35] Z. Qin, W. Sun, et al., "Linearized relative positional encoding," *Transactions on Machine Learning Research*, 2023.

[36] Z. Wang, K. Chen, et al., "POP909: A Pop-song Dataset for Music Arrangement Generation," *Conference of the International Society for Music Information Retrieval (ISMIR)*, 2020.

[37] S. Dai, H. Zhang, et al., "Automatic Analysis and Influence of Hierarchical Structure on Melody, Rhythm and Harmony in Popular Music," *Joint Conference on AI Music Creativity (AIMC)*, 2020.

[38] Y. Bengio, J. Louradour, et al., "Curriculum learning," *International Conference on Machine Learning (ICML)*, 2009.

[39] J.-C. Wang, J. B. L. Smith, et al., "Supervised metric learning for music structure features," *Conference of the International Society for Music Information Retrieval (ISMIR)*, 2021.

[40] S.-L. Wu and Y.-H. Yang, "Musemorphose: Full-song and fine-grained piano music style transfer with one transformer vae," *IEEE/ACM Transactions on Audio, Speech, & Language Processing (TASLP)*, vol. 31, pp. 1953–1967, 2023, ISSN: 2329-9290.

[41] B. Haki, M. Nieto, et al., "Real-time drum accompaniment using transformer architecture," *Joint Conference on AI Music Creativity (AIMC)*, 2022.

[42] D. S. Moore and S. Kirkland, *The Basic Practice of Statistics*, 2nd. W. H. Freeman & Co., 1999.

[43] H. Levene, "Robust tests for equality of variances," *Contributions to probability and statistics. Essays in honor of Harold Hotelling*, pp. 279–292, 1961.

[44] B. L. Welch, "The generalization of 'student's' problem when several different population variances are involved," *Biometrika*, vol. 34, no. 1/2, pp. 28–35, 1947.

[45] A. Favero, A. Sclocchi, et al., "How compositional generalization and creativity improve as diffusion models are trained," *arXiv*, 2025. [Online]. Available: https://arxiv.org/abs/2502.12089

[46] K. Mei, H. Talebi, et al., "The power of context: How multimodality improves image super-resolution," *arXiv*, 2025. [Online]. Available: https://arxiv.org/abs/2503.14503